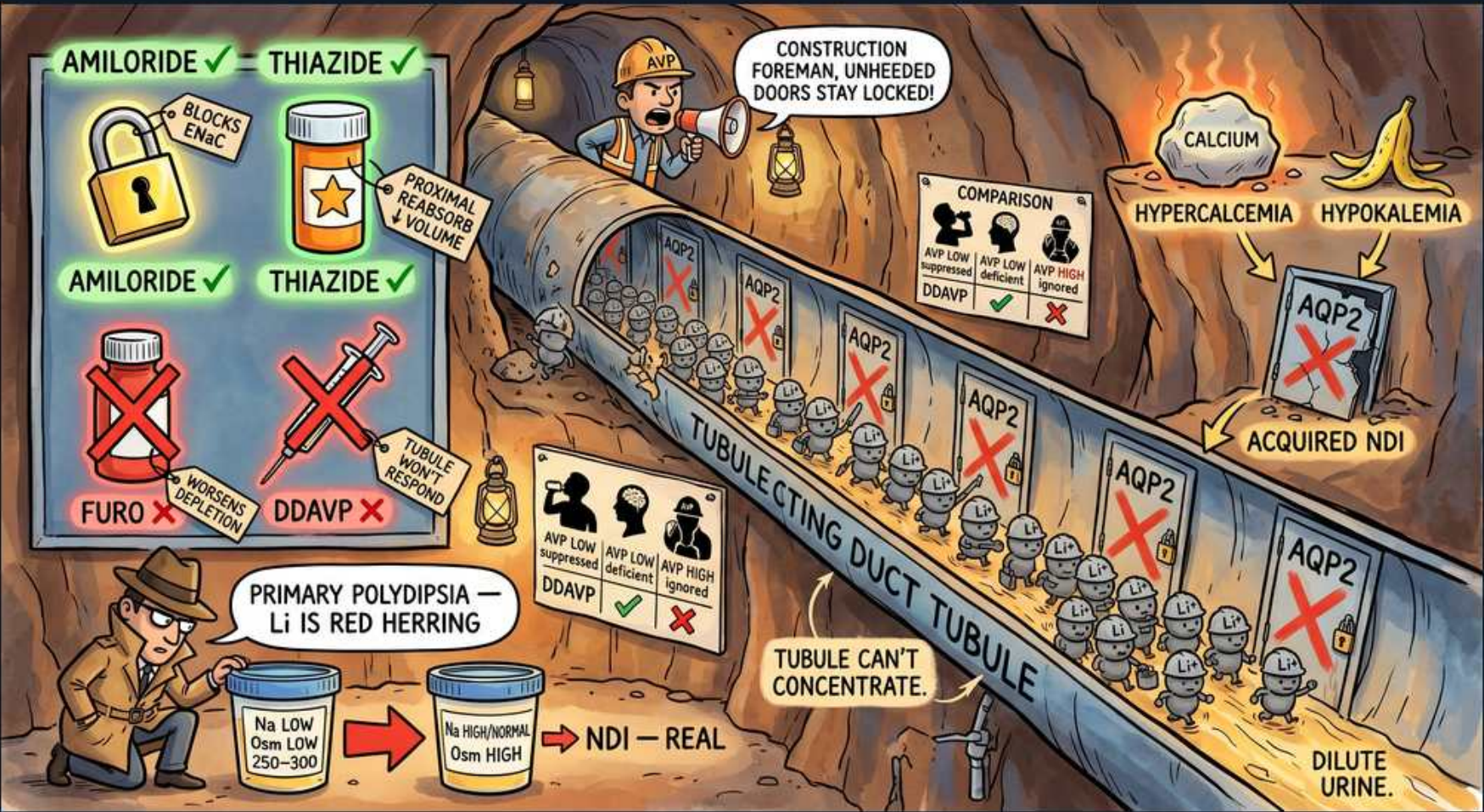


Nephrogenic DI: The Locked Collecting Duct



1. AVP foreman, doors stay locked

WHAT YOU SEE

Construction foreman 'AVP, doors stay locked'

WHAT IT ENCODES

In nephrogenic DI the collecting duct fails to respond to AVP (ADH): the aquaporin-2 'doors' don't insert despite present hormone. Result is impaired water reabsorption and inappropriately dilute urine despite hyperosmolar plasma.

BOARD TRAP

Treating NDI with DDAVP — the defect is renal resistance, so vasopressin/DDAVP fails to concentrate the urine.

2. AQP2 channels stay shut

WHAT YOU SEE

Row of locked 'AQP2' doors along the duct

WHAT IT ENCODES

AVP normally binds V2 receptors → cAMP → insertion of aquaporin-2 water channels in the apical collecting duct membrane. In NDI this signaling/insertion is blocked, so water can't be reabsorbed and urine stays dilute.

BOARD TRAP

Localizing the lesion to the hypothalamus/pituitary — NDI is a collecting-duct (kidney) defect, not a central AVP deficiency.

3. Lithium is the classic cause

WHAT YOU SEE

Detective: 'PRIMARY POLYDIPSIA — Li IS RED HERRING'

WHAT IT ENCODES

Lithium enters principal cells via ENaC and disrupts AVP-stimulated aquaporin-2 trafficking, the most common acquired NDI. Other causes: hypercalcemia, hypokalemia, chronic obstruction, demeclocycline.

BOARD TRAP

Blaming primary polydipsia for the polyuria in a lithium patient — lithium-induced NDI is the real cause; polydipsia is the red herring.

4. Amiloride blocks ENaC (lithium NDI Rx)

WHAT YOU SEE

Padlock 'AMILORIDE BLOCKS ENaC'

WHAT IT ENCODES

Amiloride blocks the ENaC channel that lithium uses to enter principal cells, reducing intracellular lithium and partially reversing lithium-induced NDI while allowing lithium to be continued.

BOARD TRAP

Stopping lithium as the only option — amiloride can mitigate lithium NDI by blocking its ENaC entry route.

5. Thiazide: paradoxical antidiuresis

WHAT YOU SEE

Pill bottle 'THIAZIDE — proximal reabsorb / ↓ volume'

WHAT IT ENCODES

Thiazides paradoxically reduce urine output in NDI by inducing mild volume contraction → enhanced proximal Na/water reabsorption → less water delivered distally. First-line symptomatic therapy along with low-salt, low-protein diet.

BOARD TRAP

Avoiding diuretics in a polyuric patient — thiazides actually decrease urine volume in NDI via proximal reabsorption.

6. Furosemide worsens depletion

WHAT YOU SEE

Crossed-out 'FURO — worsens depletion'

WHAT IT ENCODES

Loop diuretics impair the medullary concentration gradient and worsen volume/electrolyte depletion, aggravating NDI polyuria. They are the wrong diuretic class here.

BOARD TRAP

Reaching for furosemide to 'treat' polyuria — it worsens NDI; the thiazide is the counterintuitive correct choice.

7. DDAVP fails in NDI

WHAT YOU SEE

Crossed-out 'DDAVP — tubule won't respond'

WHAT IT ENCODES

DDAVP (desmopressin) is a V2 agonist that works in CENTRAL DI but fails in nephrogenic DI because the tubule cannot respond. The DDAVP response is the key test separating central from nephrogenic DI.

BOARD TRAP

A normal urine-concentrating response to DDAVP — that excludes NDI and indicates central DI instead.

8. Hypercalcemia causes acquired NDI

WHAT YOU SEE

'CALCIUM → HYPERCALCEMIA → AQP2'

WHAT IT ENCODES

Hypercalcemia activates the calcium-sensing receptor in the collecting duct, downregulating aquaporin-2 and causing reversible NDI with polyuria. Correcting calcium reverses the concentrating defect.

BOARD TRAP

Missing hypercalcemia as a reversible NDI trigger — work up Ca (and K) before labeling NDI idiopathic.

9. Hypokalemia causes acquired NDI

WHAT YOU SEE

'HYPOKALEMIA → AQP2'

WHAT IT ENCODES

Hypokalemia also downregulates aquaporin-2 and impairs the medullary gradient, producing a reversible nephrogenic concentrating defect. Repleting potassium restores urinary concentration.

BOARD TRAP

Overlooking low K as a cause of polyuria — correcting hypokalemia can resolve the NDI.

10. Acquired NDI label

WHAT YOU SEE

'ACQUIRED NDI' sign over locked AQP2

WHAT IT ENCODES

Acquired NDI (lithium, hypercalcemia, hypokalemia, obstruction, demeclocycline) is far more common than congenital (V2 receptor or AQP2 gene mutations). Identify and reverse the offending cause first.

BOARD TRAP

Assuming a genetic cause in an adult — acquired/drug-electrolyte NDI dominates and is often reversible.

11. Water deprivation: high Osm, no concentration

WHAT YOU SEE

Beakers 'Na HIGH/NORMAL, Osm HIGH → NDI REAL'

WHAT IT ENCODES

In DI a water deprivation test fails to concentrate urine (urine stays dilute, Osm low) while serum Na/Osm rise. Then DDAVP distinguishes: no response = nephrogenic; concentrates = central.

BOARD TRAP

Confusing primary polydipsia (low plasma Osm, low Na) with DI — in DI plasma Osm/Na are HIGH; polydipsia gives LOW values.

12. Tubule can't concentrate → dilute urine

WHAT YOU SEE

'TUBULE CAN'T CONCENTRATE / DILUTE URINE'

WHAT IT ENCODES

The net result of failed AVP action is persistently dilute urine (low urine osmolality, often <300) despite a hyperosmolar plasma — the hallmark of diabetes insipidus, here nephrogenic in origin.

BOARD TRAP

Reading low urine osmolality as 'overhydration' — with high plasma Osm it signals a concentrating defect (DI), not water excess.
